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Big Data Performance Analysis on a Hadoop Distributed File System Based on Geometric Data Perturbation Technique

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Abstract

In this paper, we proposed a Big Data Security Analysis based on a modified perturbation technique in a Hadoop Distributed File System which is commonly used in various real time applications. This work have utilized an improved K-means clustering algorithm, which selects the initial clustering centres based on the density parameters. The poor assurance of initial centres and in order to improve the precision and packing effect of the K-means clustering computation is to be need a new method. In order to group the data of same cluster based on their similarities clustering algorithm was used. In the various clustering methods, this paper is utilizing the geometric data perturbation technique which is clustered, data are perturbed whose values are challenging to be recognized. The performance was evaluated on a sample Health Care database and the metrics under studies are memory usage, precision, recall, accuracy, F-measure, clustering time and execution time, Perturbation time, De-perturbation time. In this paper, we focused only on Perturbation Time, De-Perturbation Time and the Accuracy .The experimental result shows that the proposed approach reduces the complexity and shown to less computation time and better accuracy than that of the existing techniques.

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Keywords: Big Data, Hadoop Distributed File System, Geometric Data Perturbation, K-means Clustering

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1. Introduction

Many successful techniques on data mining, extraction of patterns and knowledge from Big Data set were used by an efficient decision making to earn the customer satisfaction [1]. The clustering techniques can be further classified into hierarchical, partition and density based clustering. The Geometric data Perturbation (GDP) [4, 5] algorithm is utilized for perturbing the clustered data; these values are difficult to be perceived.

2. Related works

N. Bin (2018) et.al has studied about the "Internet of Things" and "Big Data" have turned into a firmly related field of use of innovation [2]. The conventional K-means calculation is streamlined to make it appropriate for the requirements of Big Data RFID Internet of Things information. In light of the Hadoop cloud clustering stage, K-means clustering examination was executed.

Aparna shinde (2016) et.al has given security to delicate standard that segregate information based on recurrence [3]. Perturbation method was utilized for concealing delicate guidelines. Analysis is done on genuine grown-up dataset for various proportions. Results demonstrates that proposed work is better in keeping up the Perturbation Percentage, Individual Privacy finally smother rules while different principles are stay unaffected.

3. Existing Method

In the existing method used K-means clustering algorithm, which selects the initial clustering centres based on the density parameters. The poor assurance of initial centres and in order to improve the precision and packing effect of the K-means clustering computation to need a new method. [5]

4. Proposed Method

4.1. Problem Statement

Perturbation replaces unique qualities with new ones by trading, including noise or making artificial information. These anonymization activities decline information utility, which is spoken to by data misfortune all in all. To tackle the above issues we propose a combined approach for productive big data performance.

4.2. The proposed approach on HDFS based on GDP Technique

The foremost focus of GDP is to give security before conveying the data. The grouped information is accomplished by methods for Geometric Data Perturbation strategy (GDP), which is applied to the clustered information, which makes it's difficult to be perceived by the unapproved clients. By then, the perturbed information were warehoused in the perturbed database.

4.3.1. Geometric Data Perturbation (GDP) Technique

The perturbation technique takes the data from any one of the tables to randomize the data. The Geometric data Perturbation (GDP) can be represented by the following equation:

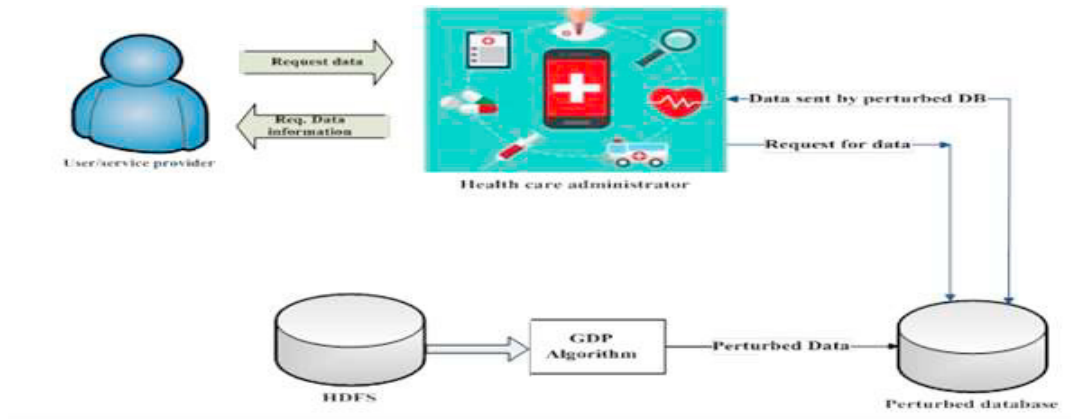


Figure 1: Proposed HDFS Based GDP framework

$$\text{GDP} = \text{TP} + \text{Vs} + \Omega$$

(1)

$$\text{Where, } \Omega = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

(2)

$T \rightarrow$ Random Projection Matrix,
 $P \rightarrow$ Original Data,
 $V_s \rightarrow$ Translational Matrix,
 $\Omega \rightarrow$ Random Gaussian Noise.

The GDP algorithm only perturbs the numerical data. After numerical transformation, the values are accessible in the P. In the event that the random matrix T is positive, we can rotate the matrix in against rotate clockwise course bearing. At that point multiply the real data matrix P and the random matrix T. In the view of the original data P, the Gaussian noise Ω is introduced. At last, we add the product of original data P and the rotation matrix T, Translation transformation matrix Vs and the random Gaussian noise. At long last, the perturbed data GDP (P) is created. These perturbed data are stored in the HDFS.

For example, the obtained numerical data is transformed into a matrix for random rotation process, consider the clustered age information of the patients $A=\{24,26,29\}$, $B=\{31,34,33,36,37\}$, $C=\{49,43,47,51\}$. Now these clustered data were rotated at 180o and the obtained data is $A=\{29,26,24\}$, $B=\{37,36,33,34,31\}$, $C=\{51,47,43,49\}$, this rotated data is multiplied with the original clustered data to get the transformed data. In the random translation transformation there is an addition of Gaussian noise in order to get the perturbed data.

5. Implementation and Analysis

We are implementing a Hadoop Distributed File System (HDFS) based on Geometric Data perturbation (GDP) technique for sensitive data. For that we are taking health care Dataset in https://www.academia.edu/17510514/Big_Data_Hadoop_In_Health_Care.

5.1. Result and Discussion

After applying the perturbation on the clustered dataset we are comparing the original data with the perturbed dataset. In Table 1 and Figure 2 the time taken by the proposed strategy for perturbation is shown in the table. We demonstrate the correlation of our proposed and existing K- means clustering algorithm.

Table 1: Time taken to Perturb Health Data

File size (MB)	10	20	30	40	50
Existing approach (MS)	5923	8115	13556	19112	26550
Proposed approach (MS)	5115	6545	11445	15855	22456

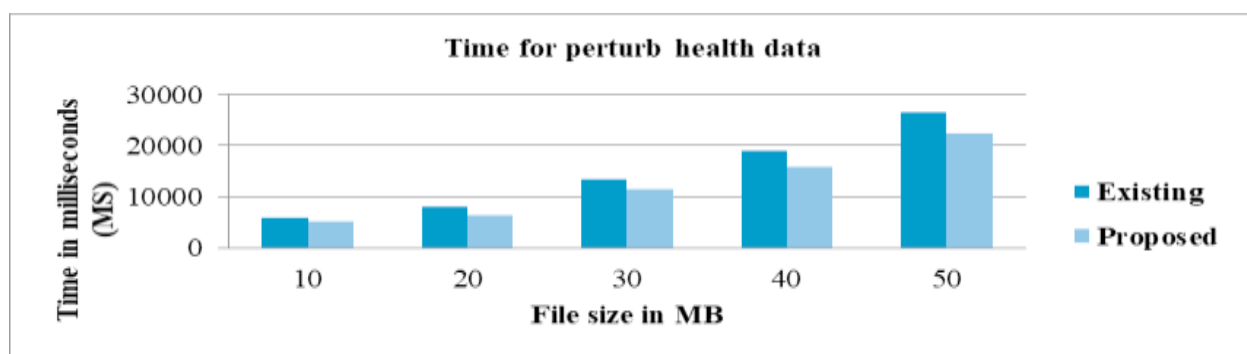


Figure 2: Comparative Analysis of Proposed and Existing method during Perturbation

Figure 2 shows the Comparative analysis of proposed and existing method during Perturbation with respect to time. For a file size of about 10 MB the time taken for perturbation by the existing approach is around 5923 MS and the time taken for perturbation by the proposed approach is around 5115 MS , this demonstrates that the proposed approach is 86% superior to that of the existing method.

Table 2: Time taken to de-perturb health data

File size (MB)	10	20	30	40	50
Existing approach (MS)	6133	8224	14177	20661	28885
Proposed approach (MS)	5333	7124	10177	17661	22885

In Table 2 and Figure 3 the Comparative investigation of proposed and existing strategy during de-perturbation/ information recovery process is appeared as for time. Here at record size of around 20 MB the time adopted by the existing approach for strategy for recovering the information is around 6133 MS and the time adopted by the proposed strategy is about 5333 MS, which is 86.95 % superior to that of the current methodology.

The time adopted by the strategies differs dependent on the size extent of the information.

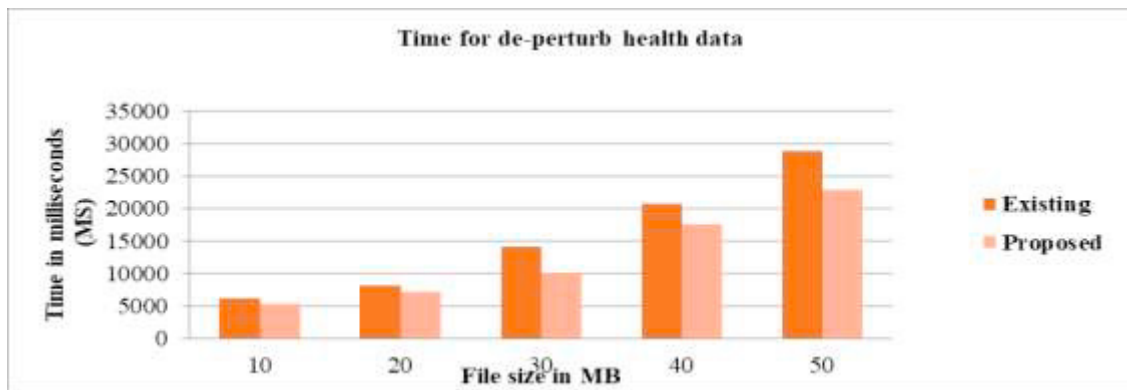


Figure 3: Comparative Analysis of Proposed and Existing method during Deperturbation.

In Table 3 and Figure 4 the accuracy achieved by proposed and existing conventional K-means clustering algorithm is shown with respect to file size. Here at file size of about 30 MB the accuracy rate by the existing approach is about 86.92 % and for the proposed approach is about 90.344 %. This suggested method ensures the entire process of clustering on time without sacrificing the accuracy of clusters.

Table 3: Accuracy

File Size (MB)	10	20	30	40	50
Existing K-Means Clustering Algorithm (%)	82.12	83.76	86.92	87.67556	91.678
Proposed GDP Algorithm (%)	84.776	86.88	90.344	91.776	93.66543

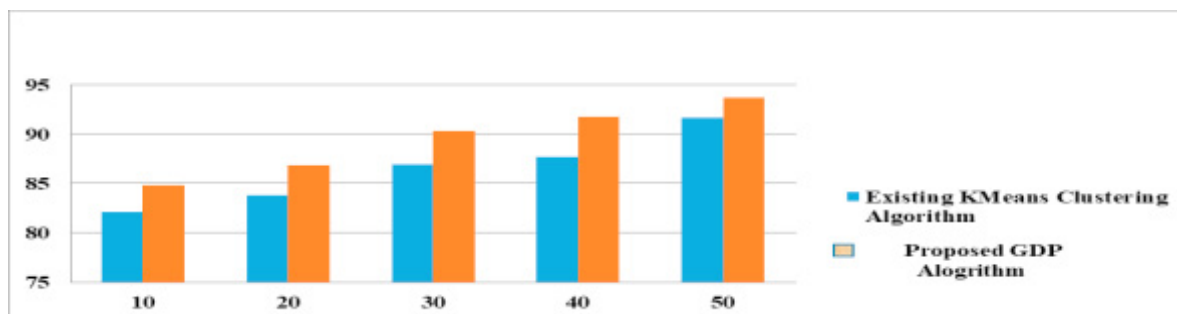


Figure 4: Accuracy comparison on proposed and existing method

6. Conclusion

The proposed algorithm were executed with several data sets and different data volume. In this GDP algorithm are utilized as the evaluation metrics like Time taken to Perturb Health Data, Time taken to De-perturb health data, Accuracy comparison on proposed and existing method to preserving the health care information exchanges by utilizing GDP. We are utilized GDP to aggravate information before redistributing to the server. In future examinations, we expect to execute therapeutic picture and sound information exchanges will be utilizing in GDP and to improve the execution of information recovery processing.

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